

The harness is part of the evaluated system

Public studies show that the coding-agent harness changes performance, cost, token use and runtime. They do not identify one harness that wins across every model and task.

Evidence captured on 12 July 2026

What the evidence supports

Harnesses are not neutral

Context, tools, retries, timeouts and caching change the system's behaviour.

The ranking changes

Pi, native harnesses, OpenCode, Cursor and research scaffolds each win somewhere.

Test the combination

The useful unit is model, harness, task and budget together.

The evidence base

This review includes 13 public evaluations. Twelve contain at least one matched-model harness comparison. Portkey is included as an efficiency diagnostic, not a quality benchmark.

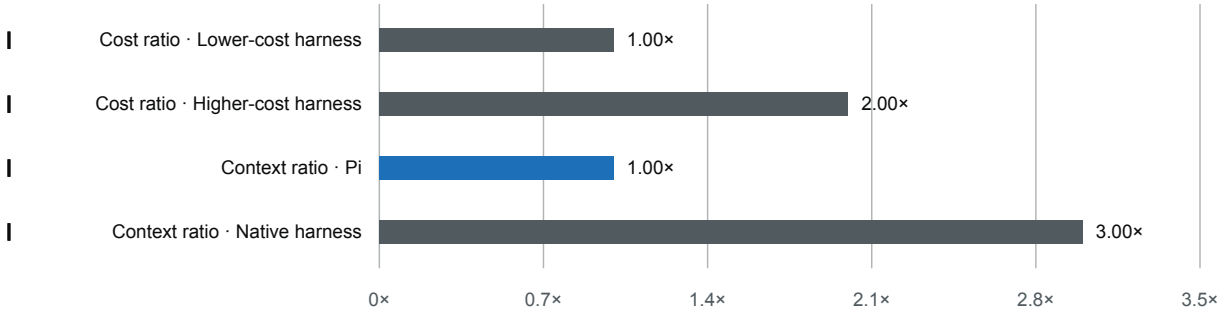
Study	Task surface	Model fixed	Includes Pi	Repeats	Cost
Databricks coding-agent benchmark	Recent human-written pull requests from a multi-million-line internal codebase	yes	yes	not clear	yes
Warden benchmark	Known-vulnerability recognition in files tied to historical Sentry vulnerabilities	yes	yes	no	yes
scBench-Long	Long biological analysis workflows	yes	yes	yes	no
OpenBench M4.5	Three repository coding tasks with executable checkers	yes	yes	yes	no

Study	Task surface	Model fixed	Includes PI	Repeats	Cost
DrugDiscoveryBench	Expert-curated drug-discovery and life-sciences workflows	yes	yes	yes	yes
Coding Agent Index	321 tasks across DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA	yes	no	yes	yes
Agent Security League	Real vulnerability-fixing tasks with hidden functional and security tests	yes	no	no	no
Vulnerability-Agent-Bench	Real vulnerability repairs in isolated containers	yes	no	no	yes
WildClawBench	Bilingual, multimodal and long-horizon agent tasks	yes	no	not clear	yes
HarnessBench	Recent bug-fix issues from 9 open-source repositories	yes	no	no	yes
CORE-Bench Hard	Scientific reproducibility tasks	yes	no	no	yes
A-CODE-CLI	Ten full-stack web development scenarios	yes	no	yes	yes
Harness Tax	Two trivial prompts routed through a gateway	no	yes	no	no
How to read harness comparisons · Download observations · Download study records					

Databricks coding-agent benchmark

Pi used about one-third as much context per turn. Some equal-quality matched runs differed by more than 2 times in cost.

Task surface	Tasks
Recent human-written pull requests from a multi-million-line internal codebase	not published
Repeated trials	Grader
not clear	Repository tests reviewed by engineers



The source reports 'more than 2 times' for cost and 'about 3 times' for context. These are directional ratios, not exact run-level values.

What this does not prove

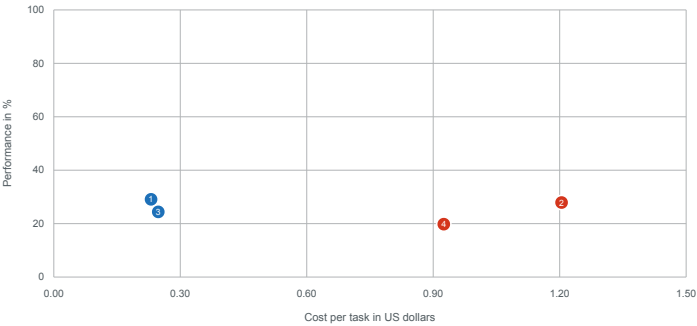
The source does not publish task count, matched rows, run-level data or uncertainty.

Databricks result source

Warden benchmark

Pi cost less and found a similar or larger share of the known corpus for the matched Claude models.

Task surface	Tasks
Known-vulnerability recognition in files tied to historical Sentry vulnerabilities	86
Repeated trials	Grader
no	Known-corpus recall



Point	Model	Harness	Performance
1	Claude Sonnet 4.6	Pi	29.1%
2	Claude Sonnet 4.6	Claude SDK	27.9%
3	Claude Opus 4.8	Pi	24.4%
4	Claude Opus 4.8	Claude SDK	19.8%

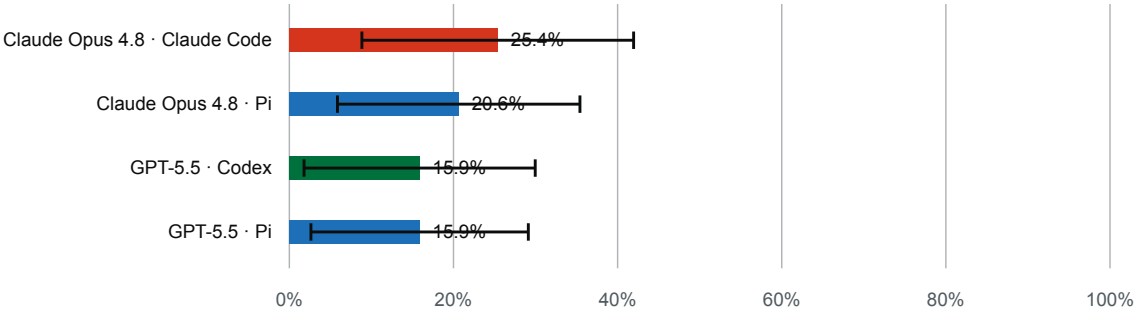
What this does not prove

The comparator is Claude SDK, not the Claude Code command-line product. The benchmark does not measure whole-repository discovery.

scBench-Long

Claude Code led Pi on Opus 4.8 while Codex and Pi tied on GPT-5.5. The reported intervals overlap widely.

Task surface	Tasks
Long biological analysis workflows	21
Repeated trials	Grader
yes	Deterministic biological graders



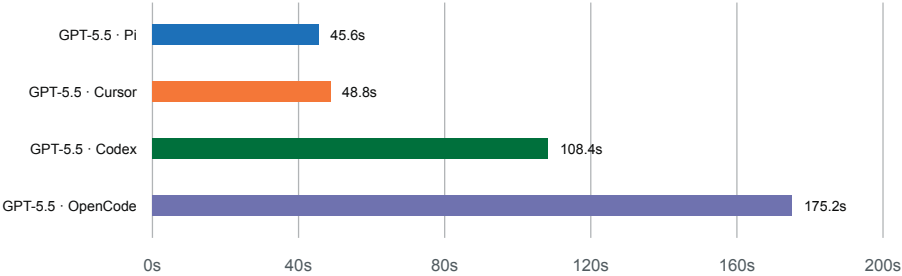
What this does not prove
The public summary does not expose cost. The confidence intervals are wide.

OpenBench M4.5

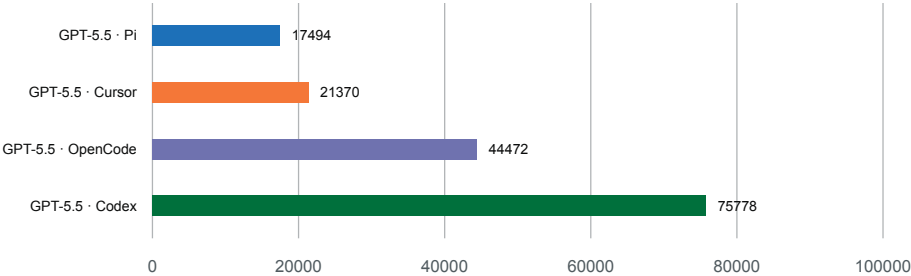
All clean harnesses passed every task. Pi and Cursor used less time and fewer tokens.

Task surface	Tasks
Three repository coding tasks with executable checkers	3
Repeated trials	Grader
yes	Executable checkers with partial-credit support

Mean wall time



Tokens per solve



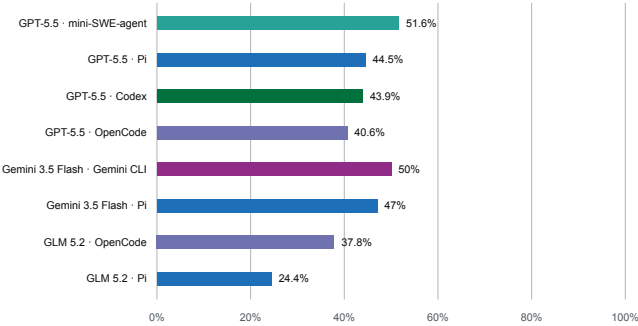
What this does not prove

The tasks were too easy to separate frontier harnesses on correctness.

DrugDiscoveryBench

GPT-5.5 performed best in mini-SWE-agent. Codex was neither the best nor worst harness for that model.

Task surface	Tasks
Expert-curated drug-discovery and life-sciences workflows	82
Repeated trials	Grader
yes	Expert-authored rubrics scored by an LLM judge



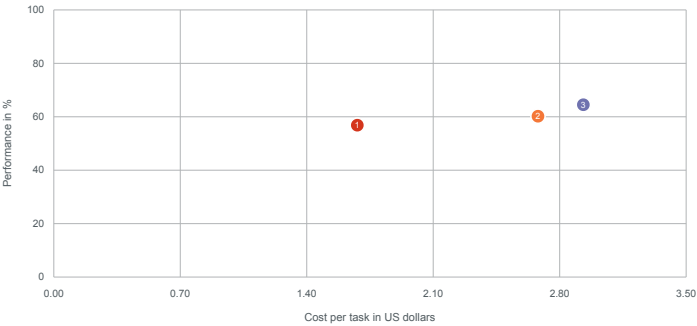
What this does not prove

The paper plots cost but does not publish a table of exact cost points. The outcome depends on an LLM judge.

Coding Agent Index

OpenCode led the captured Opus 4.7 medium comparison. Claude Code was cheaper and faster.

Task surface	Tasks
321 tasks across DeepSWE, Terminal-Bench v2 and SWE-Atlas-QnA	321
Repeated trials	Grader
yes	Program checks, test suites and rubric scoring



Point	Model	Harness	Performance
1	Claude Opus 4.7	Claude Code	56.9%
2	Claude Opus 4.7	Cursor CLI	60.2%
3	Claude Opus 4.7	OpenCode	64.5%

What this does not prove

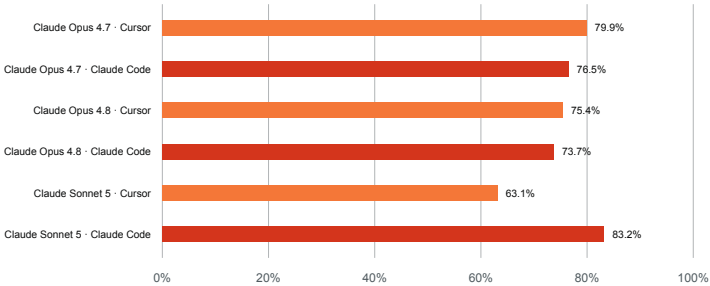
The index averages 3 different benchmark families. The live results and harness versions can change.

Agent Security League

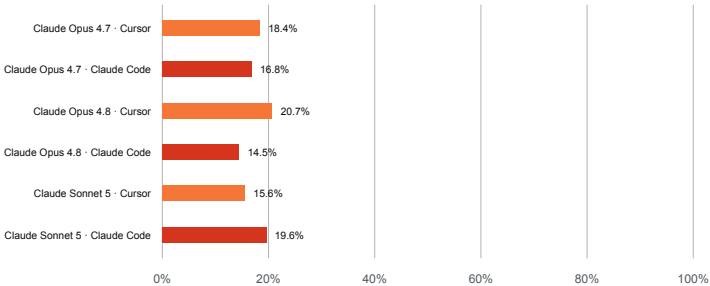
Cursor led Claude Code for Opus 4.7 and 4.8. Claude Code led Cursor for Sonnet 5.

Task surface	Tasks
Real vulnerability-fixing tasks with hidden functional and security tests	200
Repeated trials	Grader
no	Hidden functional and security tests

Functional pass rate



Security pass rate



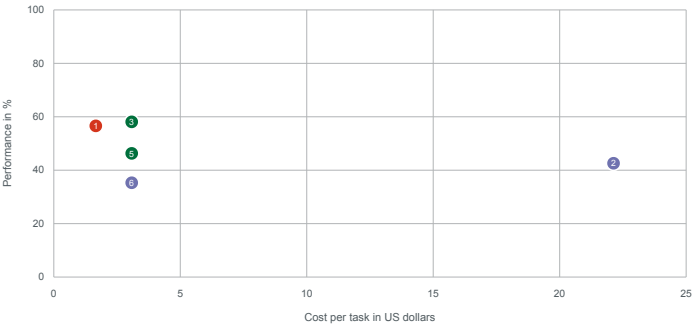
What this does not prove

There are no repeated trials or task-level costs. Timeouts explain part of the reversal.

Vulnerability-Agent-Bench

Claude Code and Codex led the corresponding OpenCode configurations on the published matched pairs.

Task surface	Tasks
Real vulnerability repairs in isolated containers	136
Repeated trials	Grader
no	Vulnerability repair checks



Point	Model	Harness	Performance
1	Claude Opus 4.6	Claude Code	56.6%
2	Claude Opus 4.6	OpenCode	42.6%
3	GPT-5.2	Codex	58.1%
4	GPT-5.2	OpenCode	46.3%
5	GPT-5.2-Codex	Codex	46.3%
6	GPT-5.2-Codex	OpenCode	35.3%

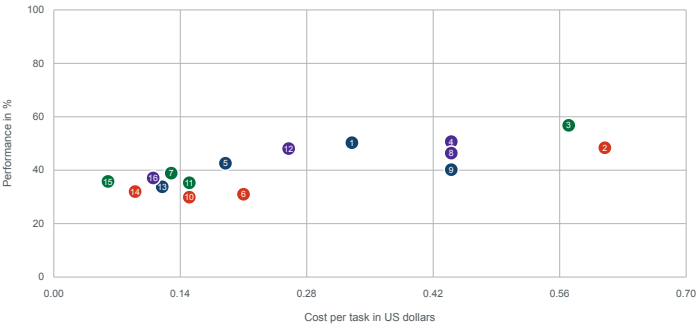
What this does not prove

The site alternates between 128 and 136 samples. Some reported cost differences need more explanation.

WildClawBench

The best harness changed with the model. The source reports double-digit score changes from switching harness.

Task surface	Tasks
Bilingual, multimodal and long-horizon agent tasks	60
Repeated trials	Grader
not clear	Deterministic checks, environment checks and model judges



Point	Model	Harness	Performance
1	GPT-5.4	OpenClaw	50.3%
2	GPT-5.4	Claude Code	48.4%
3	GPT-5.4	Codex	56.8%
4	GPT-5.4	Hermes Agent	50.7%
5	GLM 5	OpenClaw	42.6%
6	GLM 5	Claude Code	31%
7	GLM 5	Codex	38.9%
8	GLM 5	Hermes Agent	46.4%
9	MiMo V2 Pro	OpenClaw	40.2%
10	MiMo V2 Pro	Claude Code	29.9%
11	MiMo V2 Pro	Codex	35.3%
12	MiMo V2 Pro	Hermes Agent	48.1%
13	MiniMax M2.7	OpenClaw	33.8%
14	MiniMax M2.7	Claude Code	32%
15	MiniMax M2.7	Codex	35.8%
16	MiniMax M2.7	Hermes Agent	37.1%

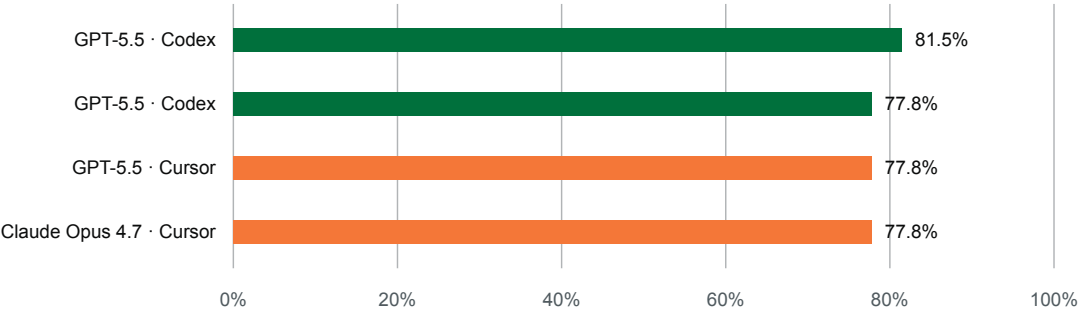
What this does not prove

Each harness image has different tools and prompt conventions. Repeat counts are not clear in the headline table.

HarnessBench

Several conditions finished within one task of the observed leader. Runtime differences were clearer than pass-rate differences.

Task surface	Tasks
Recent bug-fix issues from 9 open-source repositories	27
Repeated trials	Grader
no	Hidden core and regression tests



What this does not prove

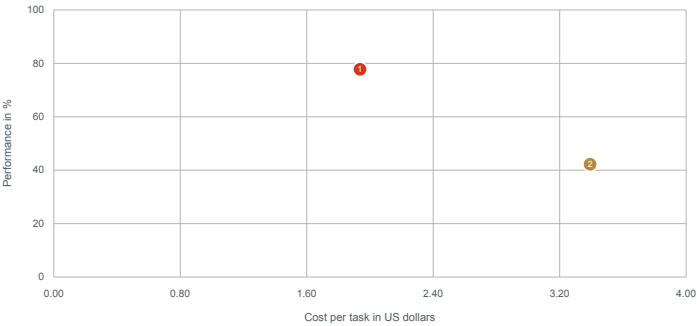
The author states that the 27-task pass-rate differences are not statistically significant.

Study 11

CORE-Bench Hard

Claude Code more than doubled the verified success of CORE-Agent for Claude Opus 4.5 while costing less.

Task surface	Tasks
Scientific reproducibility tasks	45
Repeated trials	Grader
no	Research artifact reproduction



Point	Model	Harness	Performance
1	Claude Opus 4.5	Claude Code	77.8%
2	Claude Opus 4.5	CORE-Agent	42.2%

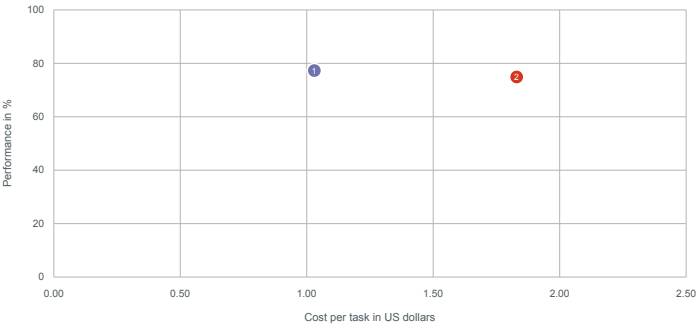
What this does not prove

There is one run per configuration. The systems differ across retrieval, tools and context policy.

A-CODE-CLI

OpenCode had a small backend lead over Claude Code and cost less in the matched Sonnet 4.6 comparison.

Task surface	Tasks
Ten full-stack web development scenarios	10
Repeated trials	Grader
yes	Backend checks and Playwright user-interface checks



Point	Model	Harness	Performance
1	Claude Sonnet 4.6	OpenCode	77.3%
2	Claude Sonnet 4.6	Claude Code	74.9%

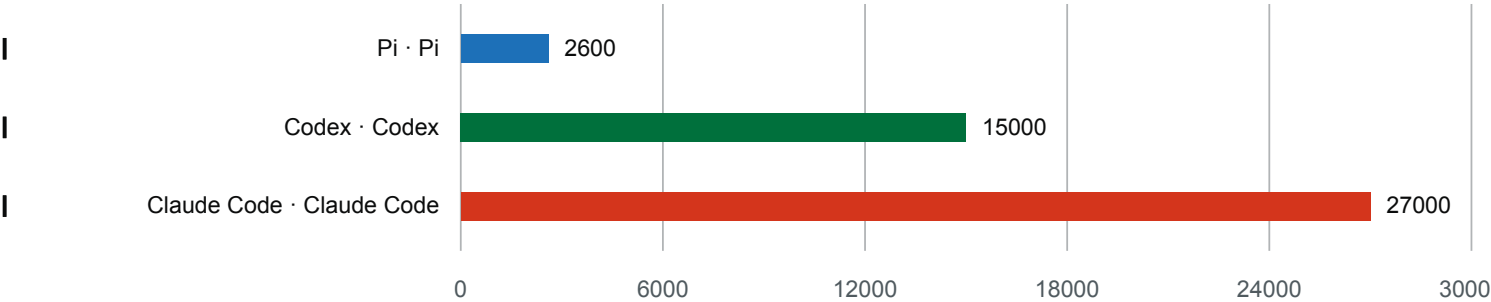
What this does not prove

The benchmark has only 10 scenarios. Only one example task is public.

Harness Tax

The fixed input payload ranged from about 2,600 tokens for Pi to 27,000 for Claude Code.

Task surface	Tasks
Two trivial prompts routed through a gateway	2
Repeated trials	Grader
no	Input-token capture



This diagnostic measures fixed input overhead. It does not measure coding quality.

What this does not prove

This is an overhead diagnostic. It does not measure coding quality or full-session economics.

Portkey result source

Data and source access

External datasets remain under their publishers' terms. This repository stores derived observations and links to the original data.

Source	Access	Licence note	Links
Benchmarking coding agents on Databricks' multi-million-line codebase	No public task-level dataset found	Not stated for benchmark data	Results
Warden benchmark	Public aggregate table	Not stated for benchmark results	Results · Data
scBench-Long	Public JSON summary and public trajectories	Check the linked benchmark repository and dataset terms	Results · Data
OpenBench	Public repository with committed data for several milestones	MIT	Results · Data
DrugDiscoveryBench	Tasks and code are public; rubrics and reference answers are gated on Hugging Face	MIT for repository code and tasks; separate terms apply to bundled data and gated material	Results · Data
Artificial Analysis Coding Agent Index	Live public result page and methodology	Not stated for result data	Results
Endor Labs Agent Security League	Public leaderboard and articles	Not stated for benchmark data	Results
Vulnerability-Agent-Bench	Public explorer and linked Weights and Biases project	Not stated for benchmark data	Results · Data
WildClawBench	Public code, tasks, runtime images and dataset	Check the repository and dataset cards	Results · Data
HarnessBench	Public code and results	Check the repository licence	Results · Data
CORE-Bench Hard	Public harness, leaderboard and encrypted traces	Check the repository licence	Results · Data
A-CODE-CLI	Public report and one example task	Check the example repository licence	Results · Data
The harness tax	Public article with aggregate values	Not stated for request logs	Results